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Questions

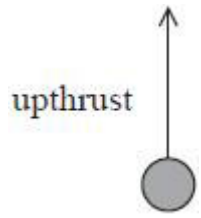
Q1.

A spherical polystyrene bead is immersed in oil. The bead has diameter 4.00×10^{-3} m. The bead is released and moves upwards through the oil at a constant velocity.

(a) Complete the free body force diagram below to show all the forces acting on the polystyrene bead.

(2)

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(b) Show that the upthrust the oil exerts on the bead is about 3.1×10^{-4} N.

density of oil = 930 kg m^{-3}

(3)

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(c) Stokes' law shows how the viscous drag on a sphere is related to its velocity through a fluid.

Stokes' law is only valid if the bead is moving sufficiently slowly through the oil.

(i) State the reason for this condition.

(1)

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(ii) For Stokes' law to be valid the speed of the bead through the oil must be less than v_R, where

$$v_R = \frac{10 \times \text{viscosity of oil}}{\text{density of oil} \times \text{diameter of the bead}}$$

Deduce whether Stokes' law can be applied to this bead.

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viscosity of oil = 4.90×10^{-2} Pa s
weight of polystyrene bead = 1.05×10^{-5} N

(5)

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(Total for question = 11 marks)

Q2.

A small, spherical air bubble moves upwards in a glass of water. The drag force on the bubble can be calculated using Stokes' law.

(a) State the condition needed for Stokes' law to apply to the bubble.

(1)

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(b) The bubble moves upwards at a constant velocity.

The volume of the bubble is 5.3×10^{-11} m^3.

(i) Show that the upthrust on the bubble is about 5×10^{-7} N.
density of water = 998 kg m^{-3}

(2)

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(ii) The bubble moves upwards at a constant velocity of 0.035 m s^{-1}.
Deduce whether Stokes' law applies to the bubble.
You may ignore the weight of the bubble.
viscosity of water = 9.5×10^{-4} Pa s

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(Total for question = 7 marks)

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Q3.

An object of mass 35 kg fell from a boat to the seabed.

(a) The object reached terminal velocity as it fell.

(i) Show that the drag force acting on the object at terminal velocity was about 200 N.
volume of object = 1.60×10^{-2} m^3
density of seawater = 1.03×10^3 kg m^{-3}

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(ii) The drag force D on the object obeyed the formula

$$D = kv^2$$

where v is the speed of the object.
Determine the terminal velocity of the object.
k = 2.2 N s^2 m^{-2}

(2)

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Terminal velocity = ...

(iii) Give **two** reasons why Stokes' law could **not** be used to calculate the terminal velocity of the object.

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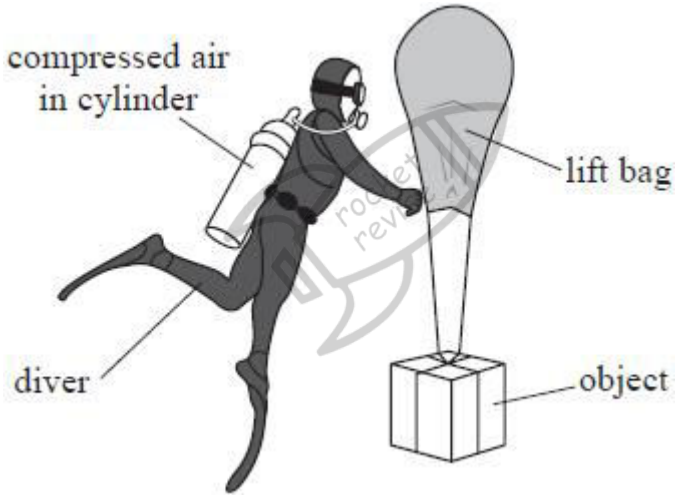
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(b) To lift the heavy object from the seabed, a diver used a 'lift bag'.

The diver used compressed air from a cylinder to fill the lift bag, as shown.



When released, the lift bag and object accelerated upwards until they reached a maximum velocity. Explain why the lift bag and object reached a maximum velocity.

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(Total for question = 12 marks)

Q4.

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Stokes' law can be used to determine the frictional force on an object moving through a fluid.

To which of the following would Stokes' law best apply?

☒ **A** A large sphere moving quickly through a fluid.

☒ **B** A large sphere moving slowly through a fluid.

☒ **C** A small sphere moving quickly through a fluid.

☒ **D** A small sphere moving slowly through a fluid.

(Total for question = 1 mark)

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Q5.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A ball bearing falls at terminal velocity through a liquid.

The temperature of the liquid increases and the ball bearing falls with a greater terminal velocity.

Which row of the table is correct as the temperature of the liquid increases?

	Viscous drag on ball bearing	Viscosity of liquid
☒ **A**	constant	increases
☒ **B**	constant	decreases
☒ **C**	decreases	increases
☒ **D**	decreases	decreases

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(Total for question = 1 mark)

Q6.

Water is dropped into a container of oil. The water forms small spherical droplets that move slowly downwards.

(a) A droplet moves downwards at a constant speed. The flow of oil around the droplet is laminar.

(i) State what is meant by laminar flow.

(1)

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(ii) State the condition necessary for the speed of the droplet to be constant.

(1)

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(b) A spherical droplet has a volume of 3.35×10^{-8} m^3.

(i) Calculate the weight of the droplet.
density of water = 1.00×10^3 kg m^{-3}

(3)

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Weight of droplet = ...

(ii) Show that the upthrust on the water droplet when it's completely submerged in oil is about 3×10^{-4} N.
density of oil = 0.94×10^3 kg m^{-3}

(2)

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(iii) Calculate the terminal velocity of this water droplet in the oil.
viscosity of oil = 0.11 Pa s

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Terminal velocity = ...

(Total for question = 11 marks)

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Q7.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A sample of sea water is collected using a beaker. The sample contains some particles of sand which settle at the bottom of the beaker.

Which of the following would result in a decrease in the time taken for the sand to settle?

- ☒ **A** smaller particles of sand
- ☒ **B** lower temperature of the sea water
- ☒ **C** smaller terminal velocity of sand particles
- ☒ **D** lower viscosity of the sea water

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(Total for question = 1 mark)

Q8.

Stokes' law can be used to determine the magnitude of the viscous drag for small, spherical objects moving through a fluid.

(a) State one other condition that must be met in order for Stokes' law to apply to the moving object.

(1)

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(b) A sphere falls through water at a constant speed of 0.50 m s^{-1}.

Assess whether Stokes' law can be applied to the falling sphere.
You should include calculations in your answer.
diameter of ball bearing = 6.0×10^{-3} m
mass of steel ball bearing = 9.1×10^{-4} kg
upthrust on ball bearing = 1.1×10^{-3} N
viscosity of water = 8.9×10^{-4} Pa s

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(Total for question = 6 marks)

Q9.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of the following best describes the situations in which Stokes' Law applies for objects moving in a fluid?

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- ☒ **A** all spherical objects moving at low speeds
- ☒ **B** all spherical objects in a fluid with low viscosity
- ☒ **C** small spherical objects moving at low speeds
- ☒ **D** small spherical objects in a fluid with low viscosity

(Total for question = 1 mark)

Q10.

A small sphere is moving horizontally through a viscous liquid.

(a) Stokes' law can be used to calculate the drag force on an object.

State the conditions that must apply for Stokes' law to be valid.

(2)

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(b) There is a constant force of 2.3×10^{-5} N acting horizontally on the sphere.

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diameter of sphere = 4.5×10^{-3} m
viscosity of liquid = 7.1×10^{-2} Pa s
(i) At one instant, the speed of the sphere is 5.2×10^{-3} m s^{-1}.
Calculate the resultant horizontal force on the sphere.

(3)

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Resultant horizontal force = ...

(ii) Calculate the maximum speed of the sphere in the horizontal direction.

(2)

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Maximum horizontal speed = ...

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(c) A larger diameter sphere in the same liquid is acted upon by the same constant force as in (b). The liquid is at a lower temperature.

Explain the effect these changes have on the maximum speed of this sphere.

(4)

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(Total for question = 11 marks)

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Q11.

A falling sphere viscometer is a device used to determine the viscosity of a liquid. A sphere falls through a cylinder containing a liquid, and the terminal velocity of the sphere is measured.

(a) State why the temperature of the liquid should **not** increase during the measurement.

(1)

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(b) A scientist measured the terminal velocity of an aluminium sphere as it fell through glycerol.

(i) Show that the weight of the sphere is about 4.8×10^{-3} N.

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density of aluminium = 2.70 × 103 kg m^{-3}
diameter of sphere = 7.00 × 10^{-3} m

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(ii) The terminal velocity of the sphere was 4.05 × 10^{-2} m s^{-1}.
Deduce whether Stokes' law applied to the falling sphere.
density of glycerol = 1.26 × 103 kg m^{-3}
viscosity of glycerol = 9.50 × 10^{-1} Pa s

(3)

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(iii) State **one** condition that must be met for Stokes' law to apply.

(1)

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(c) A scientist used the viscometer to measure the viscosity of blood. The viscosity of blood is typically in the range 3 × 10^{-3} Pa s to 4 × 10^{-3} Pa s.

Explain why the scientist should use an aluminium sphere with a much smaller diameter than the sphere used with the glycerol. Do not include calculations in your answer.
viscosity of glycerol = 9.50 × 10^{-1} Pa s

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(Total for question = 14 marks)

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Q12.

When water vapour in the atmosphere cools it condenses, forming tiny drops of water. These drops increase in size by colliding with each other and fall back to the ground as rain.

(a) As a raindrop falls through the air it eventually reaches its terminal velocity. The upthrust on the raindrop can be considered to be negligible.

(i) Explain what is meant by the terminal velocity of the raindrop. Your answer should include a free-body force diagram for the raindrop when terminal velocity has been reached.

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(ii) The air resistance F acting on a raindrop travelling at a velocity v, can be

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determined using the expression

$$F = 0.45\rho A v^2$$

where A is the cross-sectional area of the raindrop and ρ is the density of the air.
Calculate the terminal velocity of a spherical raindrop of radius 2.0×10^{-3} m.
density of air = 1.2 kg m^{-3}
density of rainwater = 1.0×10^3 kg m^{-3}

(4)

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Terminal velocity = ...

(b) Some plants have adapted to high rainfall by having a specialised shape and waxy leaves. This allows rain to slide down a leaf and off the end as a series of drops.

(i) A drop of water slides off a leaf as shown.
Add to the diagram to show the position of the drop at regular intervals of time.
The first two positions have been drawn for you.

(2)



(ii) Water falls from a leaf at a steady rate of five drops per second.
As each drop reaches the ground, there are four drops above it in the air.
Calculate the height of the leaf from the ground. It can be assumed that the drop at the highest position has just left the leaf.

(3)

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Height of leaf = ...

(Total for question = 13 marks)

Q13.

A student investigated small ball-bearings falling through a liquid to determine the viscosity of the liquid.

(a) State the conditions needed for Stokes' law to apply to this investigation.

(2)

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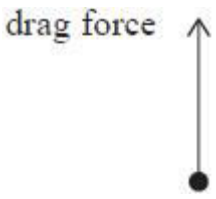
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(b) (i) Complete the free-body force diagram for a ball-bearing when falling at terminal velocity.

(2)



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(ii) The student thought that the liquid she used might be glycerol. She made measurements to determine the terminal velocity of a ball-bearing.
The terminal velocity of the ball-bearing was 0.0824 m s^{-1}.
Deduce whether the liquid she used was glycerol.
density of glycerol = 1.26×10^{3} kg m^{-3}
radius of ball-bearing = 2.30 mm
weight of ball-bearing = 4.00×10^{-3} N
viscosity of glycerol at room temperature = 0.934 Pa s

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(Total for question = 10 marks)

Q14.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

An object is moving through a fluid.

Which row of the table states the conditions which must be met for Stokes' law to apply?

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	Size of object	Shape of object	Type of flow
☐ A	any	any	turbulent
☐ B	any	spherical	turbulent
☐ C	small	spherical	laminar
☐ D	small	any	laminar

(Total for question = 1 mark)

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Q15.

A student carried out an experiment using a data logger to record the motion of a ball-bearing as it fell through a liquid.

(a) State the conditions under which Stokes' law applies to an object moving through a fluid.

(2)

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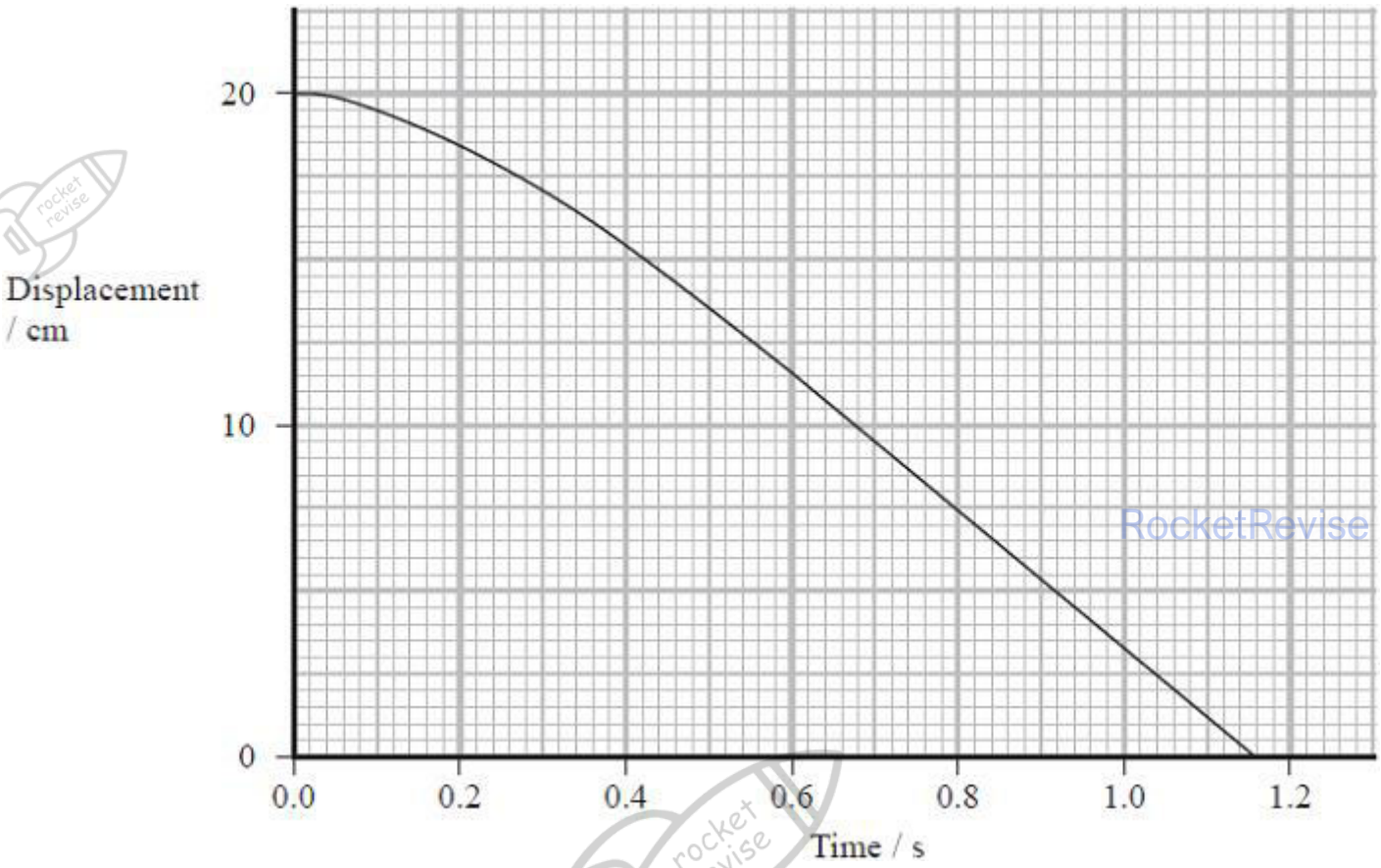
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(b) The graph obtained by the student is shown below.

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(i) Explain the shape of the graph.

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(ii) The ball-bearing had a diameter of 1.6 mm.
The difference between the upthrust on the ball-bearing and the weight of the ball-bearing is 1.45×10^{-4} N.
Calculate the viscosity of the liquid. Assume that Stokes' law applies.

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Viscosity = ... Pa s

(iii) The student heated the liquid and repeated the experiment.
The ball-bearing took less time to fall a given distance in the liquid when the liquid was at a higher temperature.
Explain why. Assume the density of the liquid is constant.

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(Total for question = 13 marks)

Q16.

A deep-sea diver is a person who swims at great depths below the surface of the sea.

(a) To help a deep-sea diver move downwards quickly, spherical masses are attached to the diver.

During a dive, one mass became unattached and fell to the bottom of the sea.
(i) State the conditions for Stokes' law to apply.

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(ii) The mass accelerated to terminal velocity as it fell to the bottom of the sea.
At terminal velocity, the mass fell 25 m in a time of 0.36 s.
There was a difference of 4.6 N between the weight of the mass and the upthrust on the mass.
The spherical mass has a volume of 45 cm^3.
Deduce whether Stokes' law applied as the mass fell through the water.
$\eta_{water} = 1.2 \times 10^{-3}$ Pa s

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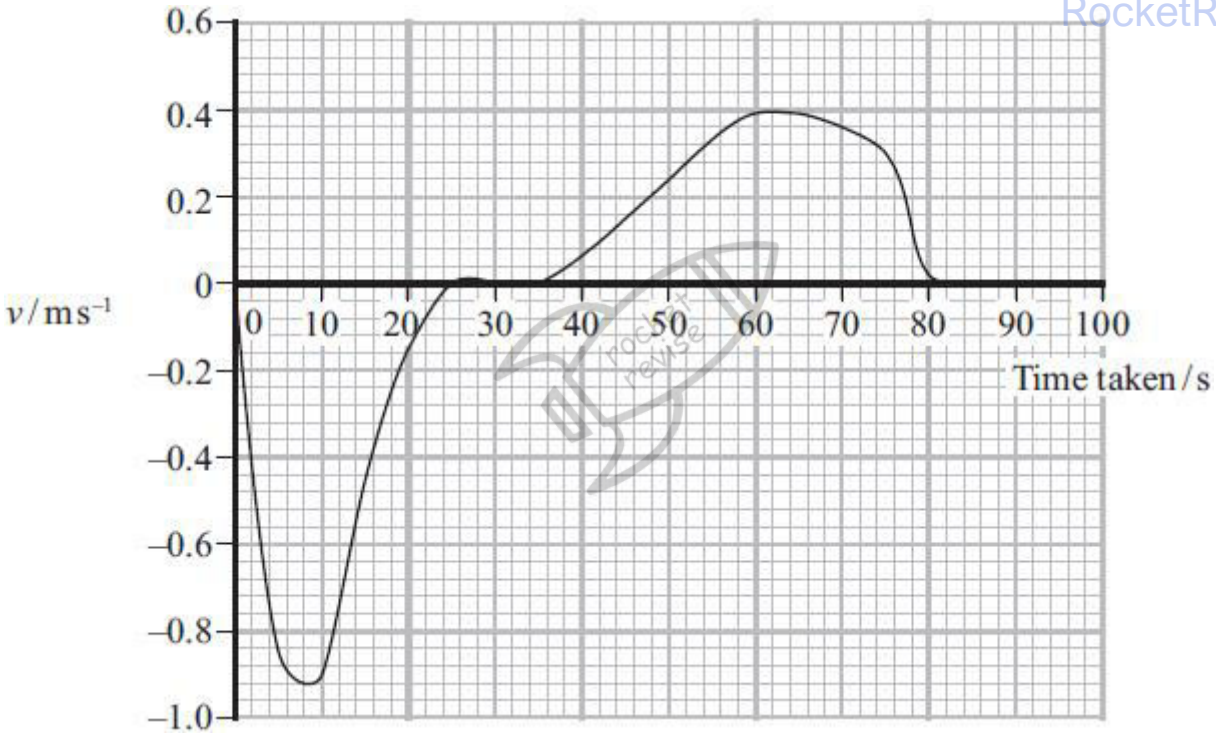
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(b) The graph shows how the vertical velocity v of the diver varied during a dive.



(i) The diver was at the surface of the sea at 0 s.
Explain how the graph could be used to show that the diver reached the surface of the sea again at 82 s.

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(ii) Determine the vertical acceleration of the diver at 70 seconds.

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Acceleration = ...

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(Total for question = 12 marks)

Q17.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A student measured the terminal velocity of different objects as they fell through a liquid. The student used the measurements and Stokes' Law to calculate the viscosity of the liquid.

For which of the following conditions does Stokes' Law apply?

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- ☐ **A** spherical objects and laminar flow
- ☐ **B** spherical objects and low viscosity
- ☐ **C** cylindrical objects and laminar flow
- ☐ **D** cylindrical objects and low viscosity

(Total for question = 1 mark)

Q18.

A grain of sand quickly reaches terminal velocity when it falls into water.

(a) State why the grain of sand reaches terminal velocity.

(1)

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(b) A spherical grain of sand has a weight of 4.3×10^{-5} N. The grain of sand displaces water with a weight of 1.5×10^{-5} N.

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The terminal velocity of the grain of sand as it falls through the water is about 0.050 m s^{-1}.
Deduce whether Stokes' law applies to the grain of sand as it falls at terminal velocity.
viscosity of water = 1.2×10^{-3} Pa s
diameter of grain of sand = 1.5×10^{-3} m

(3)

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(c) Explain how the temperature of the water affects the terminal velocity of a grain of sand if Stokes' law does apply.

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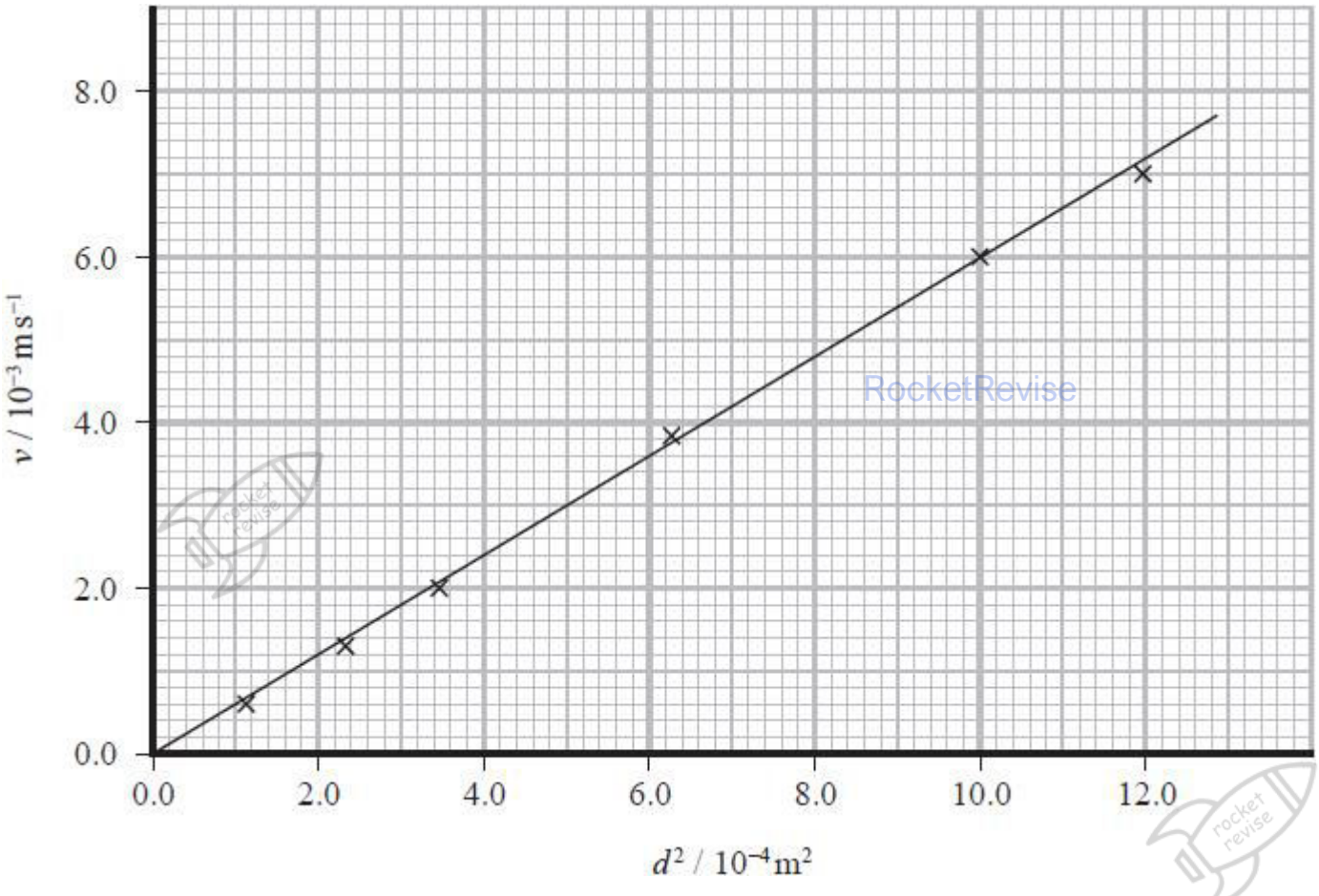
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(Total for question = 6 marks)

Q19.

A student carried out an experiment to determine the viscosity of a liquid. He measured the terminal velocities v of several different glass spheres of diameter d, as they fell through the liquid.

The student used his measurements to plot the graph of v against d^2 shown below.



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(a) Explain what is meant by terminal velocity.

(2)

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(b) A glass sphere of diameter 35 mm is travelling through the fluid at terminal velocity.

Show that the drag force on this sphere is about 0.2 N.
density of glass $\rho_g = 2.52 \times 10^3$ kg m^{-3}
density of liquid $\rho_s = 1.43 \times 10^3$ kg m^{-3}

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(c) The student reads in a textbook that if Stokes' law is obeyed

$$v = k\, d^2$$

where k is a constant.

(i) Deduce from the graph whether the flow of liquid around the spheres was laminar.

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(ii) Determine a value for k using the student's graph.

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k = ...

(iii) The constant *k* is given by

$$k = \frac{(\rho_g - \rho_s)g}{18\eta}$$

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where η is the viscosity of the liquid.
Determine a value for η.
density of glass $\rho_g = 2.52 \times 10^3$ kg m^{-3}
density of liquid $\rho_s = 1.43 \times 10^3$ kg m^{-3}

(2)

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η = ...

(Total for question = 15 marks)

Q20.

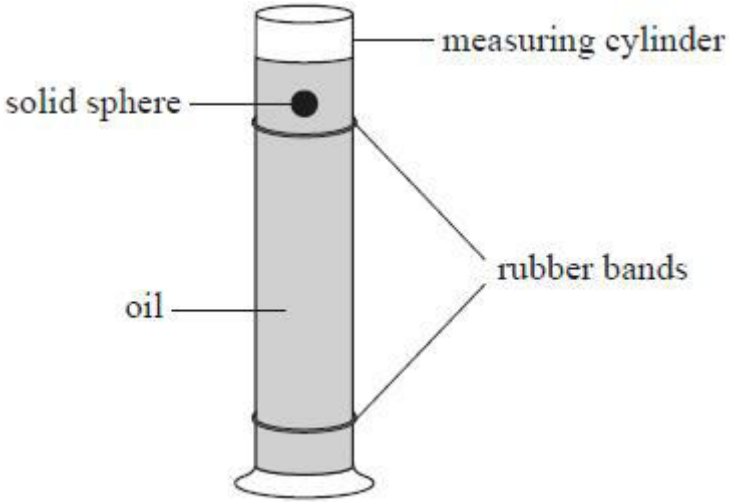
A student carried out an experiment to determine the viscosity of a sample of oil contained in a large measuring cylinder.

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A small solid sphere was released just above the surface of the oil and the terminal velocity of the sphere was determined.

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(a) Two rubber bands were placed around the cylinder as shown. When the sphere passed the top band a timer was started. The timer was stopped when the sphere passed the bottom band.

(i) Explain why the top rubber band should be placed a short distance below the surface of the oil.

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(ii) Another student suggested using an additional rubber band halfway between the top and bottom rubber bands.

Discuss the benefit of adding the third rubber band.

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(b) The student used the following equation to determine the viscosity η of the oil.

$$\frac{4}{3}\pi r^3 \rho_s g = \frac{4}{3}\pi r^3 \rho_o g + 6\pi r \eta v$$

Where
ρ_s = density of the solid sphere
ρ_o = density of the oil
r = radius of the sphere
v = terminal velocity of the sphere

(i) Which quantity is represented by $\frac{4}{3}\pi r^3 \rho_s g$?

(1)

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(ii) Which quantity is represented by $\frac{4}{3}\pi r^3 \rho_o g$?

(1)

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(iii) Which quantity is represented by $6\pi r\eta v$?

(1)

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(iv) The accepted value for the viscosity of the oil used in this experiment is 41 mPa s at 24 °C.
The student obtained a value of 36 mPa s.
Suggest a possible reason for the student obtaining a lower value for the viscosity than the accepted value.

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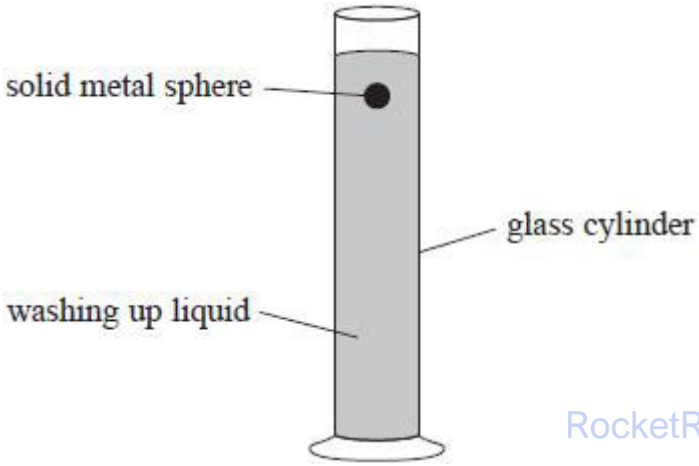
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(Total for question = 8 marks)

Q21.

A student carried out an experiment to determine the viscosity of washing up liquid using the apparatus shown.



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(a) The student released the sphere at the top of the cylinder and made measurements, using a stopwatch and metre rule, so the terminal velocity of the sphere could be determined.

(i) Describe a method that the student could use to determine an accurate value for the terminal velocity of the sphere. You may add to the diagram above.

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(ii) Explain why the use of a larger sphere would increase the percentage uncertainty in the calculated value of the terminal velocity.

(4)

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(b) (i) Complete the free-body force diagram for the sphere when travelling at terminal velocity.

(3)

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(ii) The student obtained a value of 0.16 m s^{-1} for the terminal velocity of the sphere.
Calculate the viscosity η of the washing up liquid in Pa s.
radius of sphere = 4.8×10^{-3} m
weight of sphere = 3.5×10^{-2} N
density of washing up liquid = 1.1×10^{3} kg m^{-3}

(4)

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η = ... Pa s

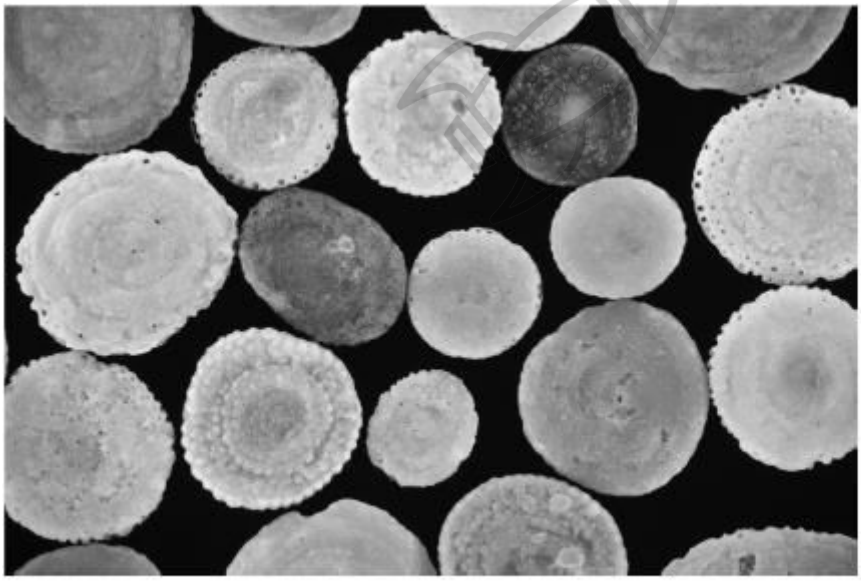
(Total for question = 16 marks)

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Q22.

Foraminifera, known as 'forams', are small sea creatures with shells.

(a) When forams die they sink slowly through the seawater to form a layer at the bottom of the sea. The photograph shows a magnified view of some typical forams.



(Source: © alex7370/Shutterstock)

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(i) A spherical foram with a mass of 1.15×10^{-8} kg is sinking at terminal velocity in seawater. The upthrust due to the seawater is 4.37×10^{-8} N. The flow of seawater around the sinking foram is laminar.

Show that the viscous drag on the foram is about 7×10^{-8} N.

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(ii) Calculate the terminal velocity of this foram.
diameter of foram = 9.2×10^{-4} m
viscosity of seawater = 1.41×10^{-3} Pa s

(2)

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Terminal velocity = ...

(b) As the forams sink deeper into the sea, the temperature of the water becomes lower.

Discuss how the terminal velocities of forams change as they sink deeper into the sea.

(4)

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(Total for question = 9 marks)

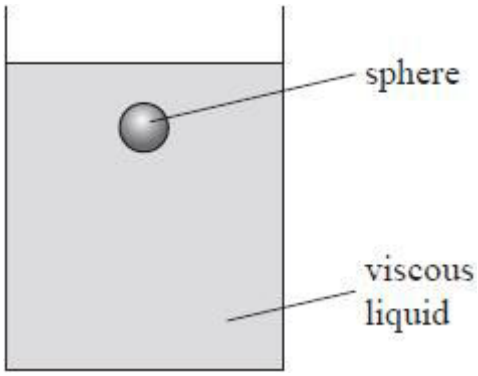
Q23.

Answer the question with a cross in the box you think is correct (☒). If you change your mind, put a line through the box (☒) and then mark your new answer with a cross (☒).

A sphere falls through a viscous liquid as shown.

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Which row of the table describes the upthrust and the viscous drag on the sphere as it accelerates downwards?

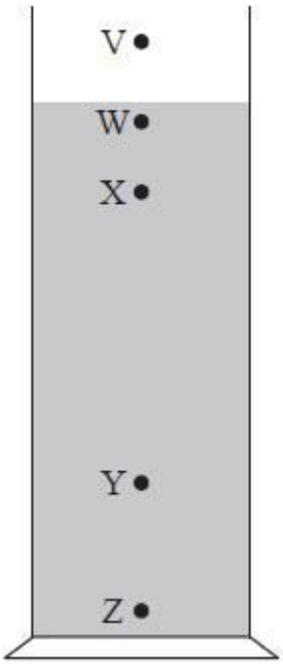
	Upthrust	Viscous drag
☒ A	increasing	increasing
☒ B	constant	increasing
☒ C	increasing	constant
☒ D	constant	constant

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(Total for question = 1 mark)

Q24.

A student determines the terminal velocity of a ball bearing as it falls through oil. He releases the ball bearing at point V and measures the time taken for it to fall a measured distance.



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Which two points should he use for the measured distance?

- ☒ **A** V and Y
- ☒ **B** W and Y
- ☒ **C** X and Y
- ☒ **D** W and Z

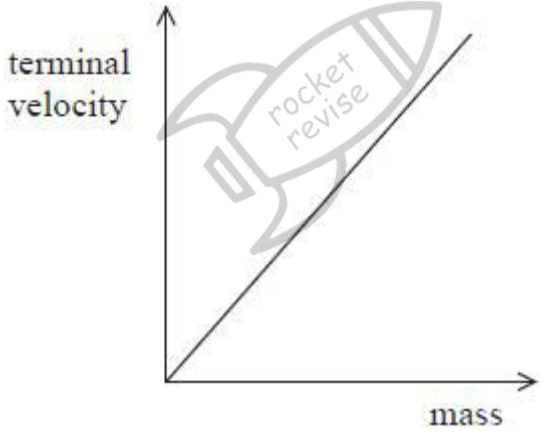
(Total for question = 1 mark)

Q25.

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Select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The graph shows how terminal velocity varies with mass for small spheres of equal diameter falling through a viscous liquid.



Which of the following describes the gradient of the graph for a liquid of greater viscosity?

- ☒ **A** a greater gradient
- ☒ **B** a smaller gradient
- ☒ **C** a variable gradient
- ☒ **D** the same gradient

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(Total for question = 1 mark)

Q26.

A student determined the viscosity of a liquid using the falling-ball method.

(a) When the ball is falling at terminal velocity the following equation applies

drag force = weight of ball – upthrust

The density of the liquid was known.

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The student used a balance and a digital calliper to make measurements on the ball.
Describe how the student could use her measurements to calculate a value for the drag force acting on the ball.

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(b) When falling through the liquid, the ball reached terminal velocity.

The flow of liquid around the ball was laminar.
Calculate the viscosity of the liquid.
terminal velocity of ball = 5.4×10^{-4} m s^{-1}
radius of ball = 0.50×10^{-2} m
drag force = 1.1×10^{-2} N

(2)

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Viscosity of liquid = ...

(Total for question = 6 marks)

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Q27.

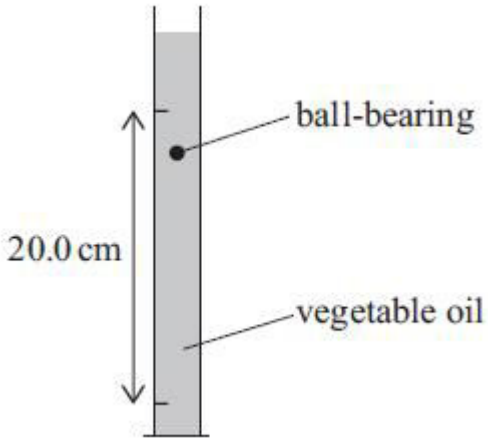
A student used a falling-ball method to determine the viscosity of vegetable oil.

He measured the diameter of a ball-bearing.

He dropped the ball-bearing into a measuring cylinder containing vegetable oil, as shown.

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The ball-bearing reached terminal velocity. The student then measured the time taken for the ball-bearing to fall 20.0 cm.

(a) Give one other measurement needed to determine the viscosity of the oil.

(1)

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(b) The diameter of the ball-bearing was 1.24 mm. The time taken to fall 20.0 cm was 1.3 s. The student calculated a value of 5.6×10^{-5} N for the viscous drag on the ball-bearing.

Calculate the viscosity of the oil.

(3)

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Viscosity = ... Pa s

(c) The student repeated the experiment. He used the same oil at a lower temperature.

Explain how using oil at a lower temperature affected the time taken for the ball-bearing to fall 20.0 cm.

(3)

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(Total for question = 7 marks)

Q28.

A student determined the terminal velocity of a metal sphere falling through a container of oil.

(a) The student released the sphere from rest in the oil.

Explain why the sphere reached terminal velocity.

(4)

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(b) The student measured the time taken for the sphere to fall through 0.45 m of the oil at terminal velocity.

(i) Calculate the viscous drag force on the sphere.
Assume the conditions for Stokes' law applied.
time taken = 7.3 s
radius of sphere = 0.90×10^{-3} m
viscosity of oil = 2.1×10^{-1} Pa s

(3)

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Viscous drag force = ...

(ii) The student repeated the experiment using a second sphere with a greater diameter.
The density of the second sphere was the same as the density of the original sphere.
The temperature of the oil was the same and the conditions for Stokes' law applied.
Explain how the time taken for the second sphere to fall through 0.45 m at terminal velocity compares with the time taken for the original sphere.

(3)

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(Total for question = 10 marks)

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